

Deploying VM in OCP-V using Redhat Gitops

Objective:

Aim of this activity is to deploy an VM as an application in OCP-V using Redhat Gitops

Prerequisites:

- Redhat Gitops operator deployed
- A public git repository containing vm.yaml and namespace.yaml

Procedure:

- Upload the vm.yaml and namespace.yaml in a git repository

vm.yaml

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: demo-vm
  namespace: vm-demo
  annotations:
    argocd.argoproj.io/sync-wave: "0"
spec:
  running: true
  template:
    metadata:
      labels:
        kubevirt.io/domain: demo-vm
    spec:
      domain:
        cpu:
          cores: 1
        resources:
          requests:
            memory: 1Gi
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
      volumes:
        - name: containerdisk
          containerDisk:
            image: kubevirt/cirros-container-disk-demo
```

Namespace.yaml

```
apiVersion: v1
kind: Namespace
metadata:
  name: vm-demo
  annotations:
    argocd.argoproj.io/sync-wave: "-1"
```

Repo url : https://github.com/JayarajVp/git_vms.git

- Now we deploy the app.yaml

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: vm-gitops
  namespace: openshift-gitops
spec:
  project: default
  source:
    repoURL: https://github.com/JayarajVp/git_vms.git
    targetRevision: HEAD
    path: .
  destination:
    server: https://kubernetes.default.svc
    namespace: vm-demo
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
```

Red Hat OpenShift GitOps fetches namespace.yaml and vm.yaml from the repository and applies them sequentially based on sync waves. The namespace is created first (argocd.argoproj.io/sync-wave: "-1"), ensuring the target environment exists before the VM (argocd.argoproj.io/sync-wave: "0") is deployed.

```
oc apply -f app.yaml
```

```
[demo-user@bastion ~]$ oc project vm-demo
Now using project "vm-demo" on server "https://api.cluster-qdw27.qdw27.sandbox2194.opentlc.com:6443".
[demo-user@bastion ~]$ oc get all
Warning: apps.openshift.io/v1 DeploymentConfig is deprecated in v4.14+, unavailable in v4.10000+
Warning: kubevirt.io/v1 VirtualMachineInstancePresets is now deprecated and will be removed in v2.
NAME                                READY    STATUS    RESTARTS   AGE
pod/virt-launcher-demo-vm-8dlwz     2/2     Running   0          50s

NAME                                AGE    PHASE    IP            NODENAME                                READY
virtualmachineinstance.kubevirt.io/demo-vm  50s    Running  10.131.0.63   ip-10-0-12-157.ec2.internal            True

NAME                                AGE    STATUS    READY
virtualmachine.kubevirt.io/demo-vm  50s    Running   True
```

- Access the web ui of Argo CD

Get the password for admin user by running the following command

```
oc get secret openshift-gitops-cluster \
  -n openshift-gitops \
  -o jsonpath='{.data.admin\.password}' | base64 -d && echo
```

Login with the user admin and password you got from running above command

Navigate to application and vm-gitops

The screenshot shows the Argo CD web UI for the 'vm-gitops' application. The sidebar on the left contains navigation links: Applications, Settings, User Info, and Documentation. The main content area displays the application's health status as 'Healthy', sync status as 'Synced to HEAD (dd7193b)', and a detailed sync history. Below this, a diagram shows the application's structure: 'vm-gitops' (Application) is linked to 'vm-demo' (Namespace), which contains 'demo-vm' (VirtualMachine). This is further linked to 'revision-start-vm-009f279e-b...' (ControllerRevision) and 'demo-vm' (VirtualMachineInstance). The sync status for each component is shown as 'Synced' with a green heart icon.